

Fundamentals of Electric Circuits

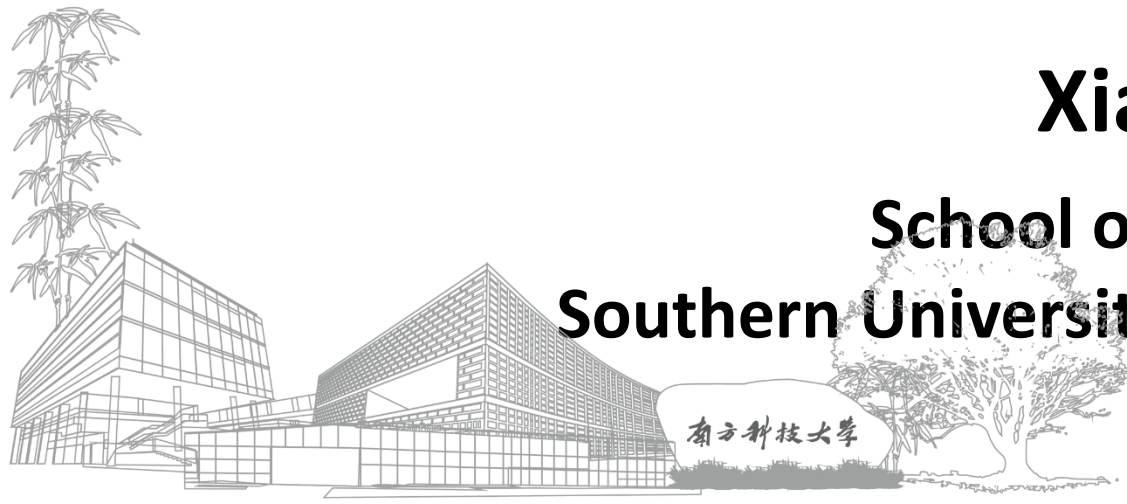
电路基础

Lecture1: Introduction and Basic Concepts

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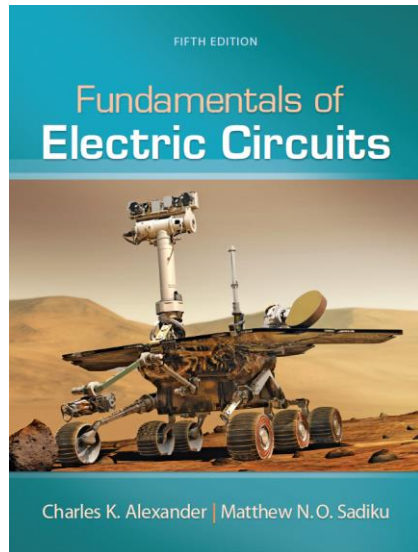


Course Overview

- **Lecture: Wednesday, 2:00 pm – 3:50 pm**
- **Blackboard is used to distribute notes, assignments, announcements**
- **Instructor: Liu Xiaolong**
- **Contact: liuxl@sustech.edu.cn**
- **TA:**
 - **Jin Xia (2957538622@qq.com)**
 - **Zheng Jing (1753340689@qq.com)**

Reference Textbooks

- Fundamentals of Electric Circuits, Charles Alexander and Matthew Sadiku, 5th Edition, McGraw-Hill Education, 2012
- 电路原理，江缉光，刘秀成，第二版，清华大学出版社，2007
- 电路，邱关源，第五版，高等教育出版社，2006



Course Assessment

- **Attendance (5%)**
- **Homework (15%)**
 - **One homework roughly every week**
 - **Essential for learning the class material**
- **Mid-Term Exam (40%)**
- **Final Exam (40%)**

Syllabus

- DC Circuits
 - Basic Concepts
 - Basic Laws
 - Methods of Analysis
 - Circuit Theorems
 - Operational Amplifiers
 - Capacitors and Inductors
 - First-Order Circuits
 - Second-Order Circuits
- AC Circuits
 - Sinusoids and Phasors
 - Sinusoidal Steady-State Analysis
 - AC Power Analysis
 - Frequency Response
 - Magnetically Coupled Circuits
- Advanced Circuit Analysis
 - Fourier Series
 - Two-Port Networks
- Focus on the analytical techniques for describing the behavior of a circuit
 - How does an electric circuit respond to a given input?
 - How do the elements and devices in the circuit interact?

Outline

- **Introduction**
- **Current**
- **Voltage**
- **Power and Energy**
- **Circuit Elements**

What is an Electric Circuit? (I)

- In electrical engineering, we are usually interested in **transferring energy** or **communicating signals** from one point to another
- To do this, we often require an interconnection of electrical components
- **An electric circuit (电路) is an interconnection of electrical components (电气元件)**

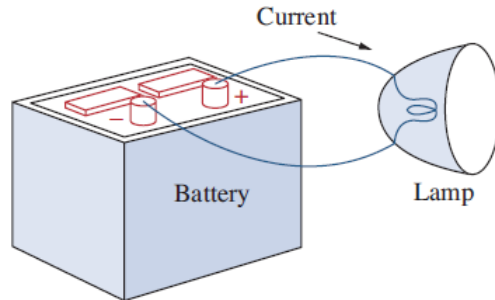


Figure 1.1
A simple electric circuit.

What is an Electric Circuit? (II)

- In electrical engineering, we are usually interested in **transferring energy** or **communicating signals** from one point to another
- To do this, we often require an interconnection of electrical components
- **An electric circuit (电路)** is an **interconnection** of **electrical components (电气元件)**
- Typical circuit or electrical components that we will see in this course:
 - Batteries or voltage sources, current sources, resistors, capacitors, inductors, magnetically coupled inductors, operational amplifiers ...

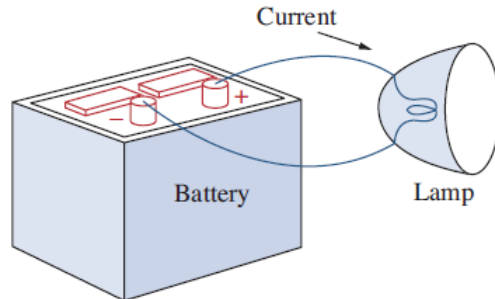


Figure 1.1
A simple electric circuit.

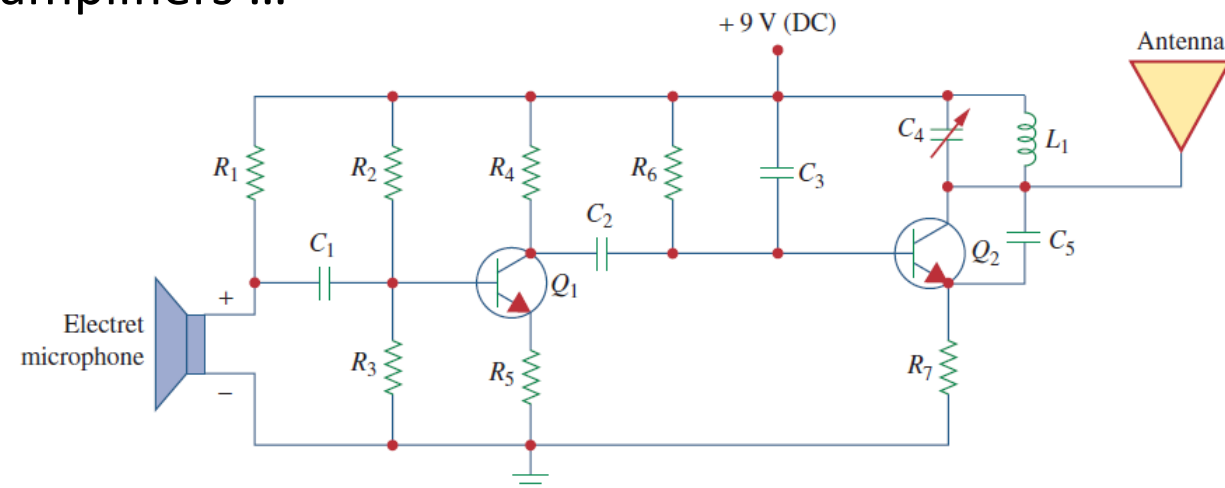
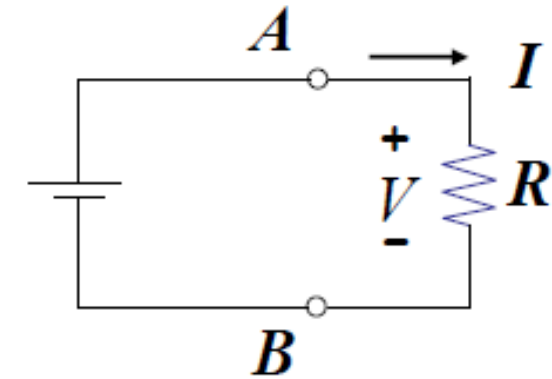
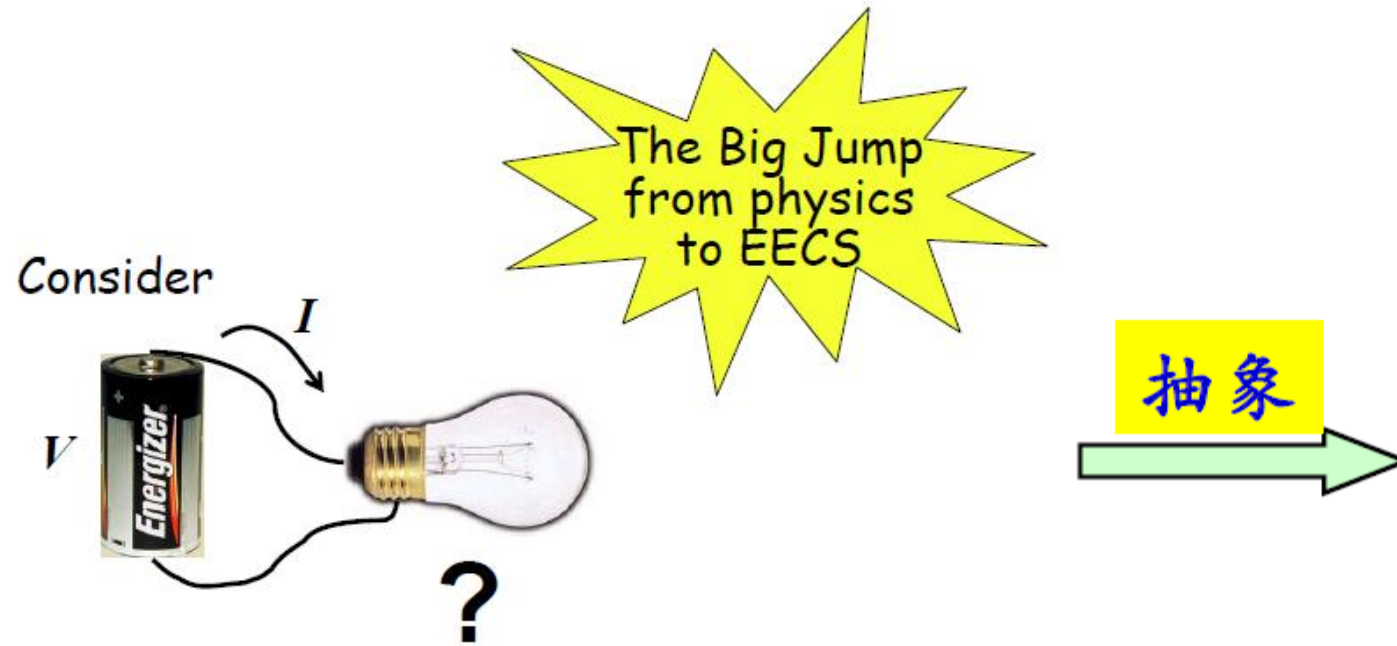


Figure 1.2
Electric circuit of a radio transmitter.

Why Should We Study Electric Circuit?

- From an academic point of view
 - The fundamentals of Electrical Engineering
 - The fundamentals of Integrated Circuits
 - The fundamentals of Automatic Control
- From a practical point of view
 - This course is a prerequisite for many advanced courses
 - Solve practical problems in study and scientific research

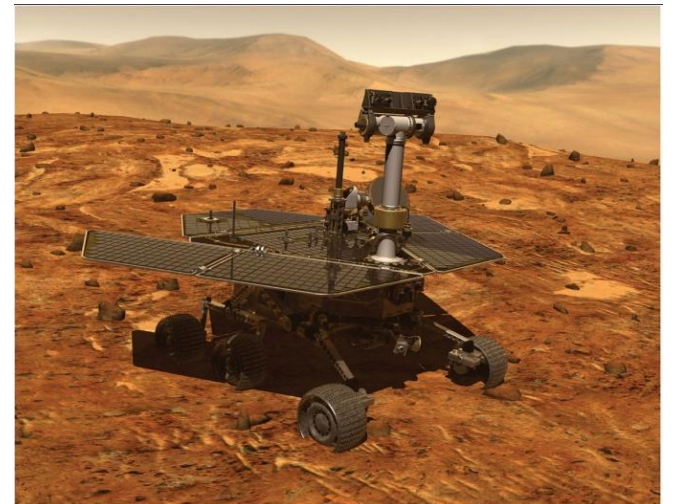
Lumped Circuit Abstraction (集总电路抽象)



R is a lumped element
abstraction for the bulb

*Suppose we wish to answer this question:
What is the current through the bulb?*

Examples of Systems Containing Electric Circuits



System of Units

- As electrical engineers, we deal with **measurable quantities**; when taking measurements, we need to use units to quantify values
- We adopt the International Systems of Units (SI for short)

TABLE 1.1

Six basic SI units and one derived unit relevant to this text.

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd
Charge	coulomb	C

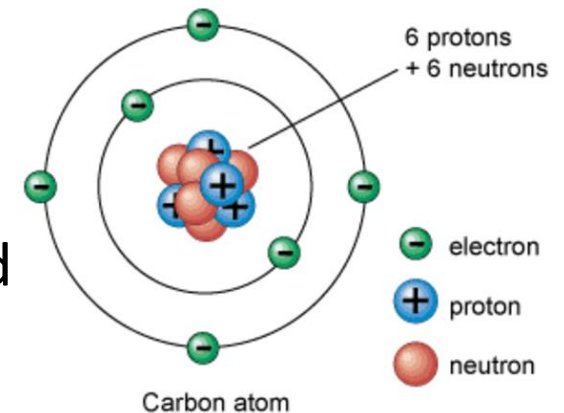
TABLE 1.2

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Review of Basic Circuit Concepts

- Electric Charge is the basis for describing all electrical phenomena
- **Charge (电荷)** is an electrical property of the atomic particles of which matter consists and is measured in **coulombs (C)**
- Inside an atom, there is negative charge on electrons, positive charge on protons and no charge on neutrons
- The charge of an electron is equal to that of a proton and is:
 - $e = 1.602 \times 10^{-19} \text{ C}$
- **The law of conservation of charge (电荷守恒定律)**
 - Charge can neither be created nor destroyed, only transferred



Electric Current (Charges in Motion!)

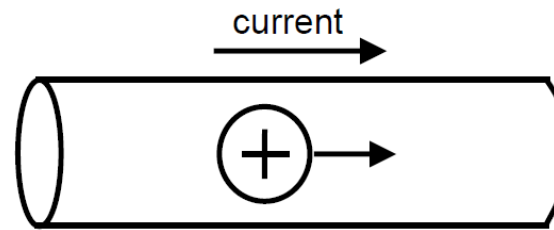
- A unique feature of electric charge is the fact that it is mobile; that is, it can be transferred from one place to another, where it can be converted to another form of energy
- **Current:** net flow of charge across any cross section of a conductor, measured in **Amperes (A)** (*Andre-Marie Ampere (1775-1836), a French mathematician and physicist*)
- Current can be thought of as the rate of change of charge

$$i(t) = \lim_{t \rightarrow 0} \frac{q}{t} = \frac{dq}{dt}$$

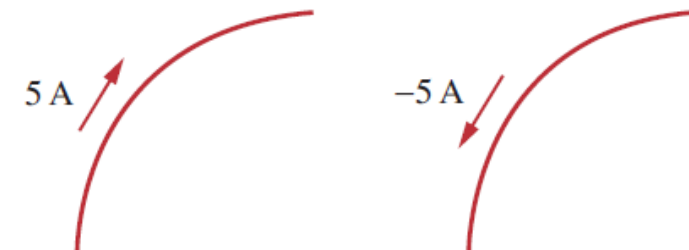
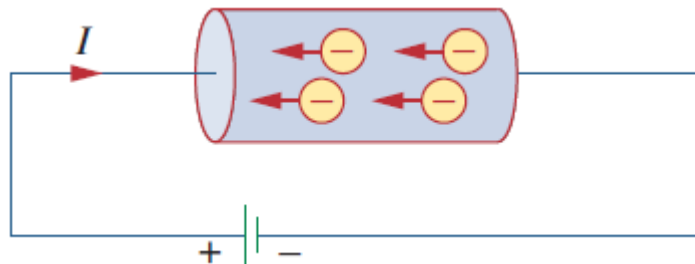


Electric Current

- Originally scientists (*in particular Benjamin Franklin (1706-1790) an American scientist and inventor*) thought that **current is only due to the movement of positive charges**. Thus, the direction of the current was considered the direction of movement of positive charges



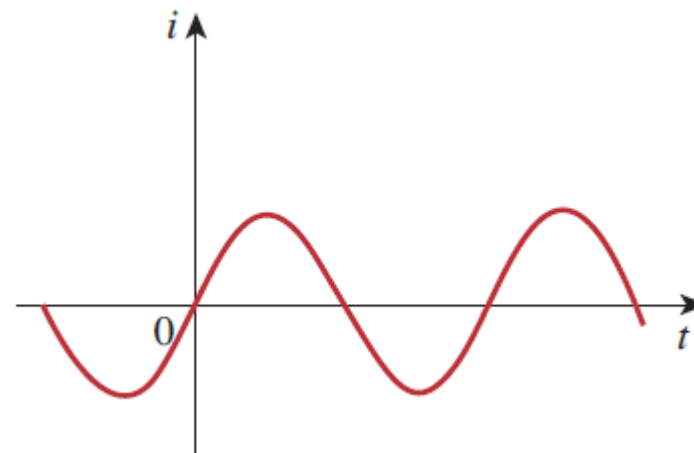
- In reality **in metallic conductors, current is due to the movement of electrons**, however, we follow the universally accepted convention that current is in the direction of positive charge movement



Two ways of showing the same current

Two Important Types of Current

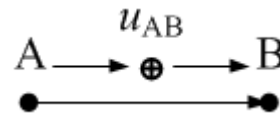
- Current can take various forms against time
- **Direct current (DC, 直流)** is a current that remains constant with time, conventionally represented by I (in **capital**). A common source of DC is a battery
- **Alternating current (AC, 交流)** is a current that varies sinusoidally with time, conventionally represented by i (in **lowercase**). Mains power of wall socket is an example of AC



Voltage

- To move the electron in a conductor in a particular direction requires some work or energy transfer. This work is performed by an external **electromotive force (电动势)**, typically due to a battery or an electric generator
- Voltage (电压)** (electromotive force, or potential) is the **energy required to move a unit charge** through a circuit element, and is measured in **Volts**
(Alessandro Antonio Volta (1745-1827) an Italian Physicist)

$$u_{AB} = \frac{dW_{AB}}{dq}$$



- Positive charge moving from a higher potential to a lower one yields energy. Moving from lower to higher potential requires energy

Voltage Polarity

- The **plus (+)** and **minus (-)** sign are used to define voltage polarity
- The assumption is that the potential of the terminal with (+) polarity is higher than the potential of the terminal with (-) polarity by the amount of **voltage drop** (电压降)
- The polarity assignment is somewhat arbitrary! Is this a scientific statement?!! What do you mean by arbitrary?!!!

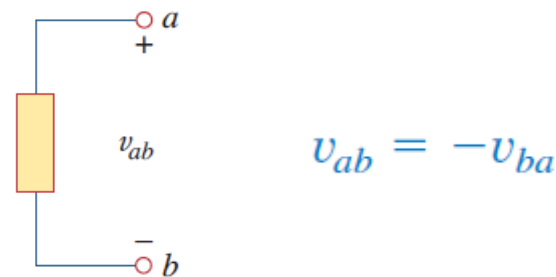


Figure 1.6
Polarity of voltage v_{ab} .

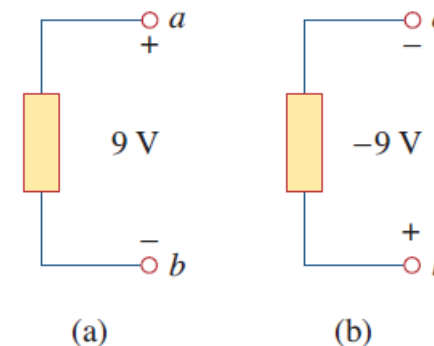
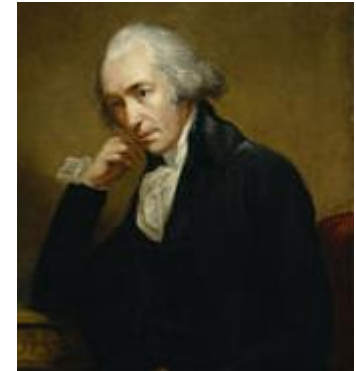
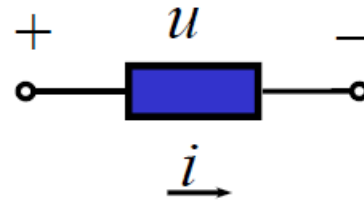


Figure 1.7
Two equivalent representations of the same voltage v_{ab} : (a) Point a is 9 V above point b ; (b) point b is -9 V above point a .

Power and Energy (I)

- For practical purposes, we need to know how much power an electric device can handle
- Power (功率)**: The rate of change of (expending or absorbing) energy per unit time, measured in **Watts (W)** (*James Watt (1736-1819) a Scottish inventor and mechanical engineer*)

$$p = \frac{dW}{dt} = \frac{dW}{dq} \frac{dq}{dt} = ui$$



- Thus, the power absorbed or supplied by an element is the product of the voltage across the element and the current through it

Power and Energy (II)

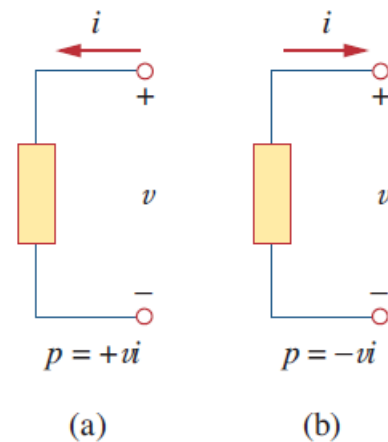
- If the power has a **+ sign**, power is being delivered to or absorbed by the element. If, on the other hand, the power has a **- sign**, power is being supplied by the element
- *How do we know when the power has a negative or a positive sign?*

Power and Energy (II)

- If the power has a **+ sign**, power is being delivered to or absorbed by the element. If, on the other hand, the power has a **- sign**, power is being supplied by the element
- *How do we know when the power has a negative or a positive sign?*
- **Passive sign convention (无源符号约定/关联参考方向)** is satisfied if the direction of current is selected such that current enters through the terminal that is more positively biased
 - When the current enters through the positive terminal of an element and $p = +v_i$. If the current enters through the negative terminal, $p = -v_i$

Power and Energy (III)

- **Passive sign convention** (无源符号约定/关联参考方向) is satisfied if the direction of current is selected such that current enters through the terminal that is more positively biased
 - When the current enters through the positive terminal of an element and $p = +vi$.
If the current enters through the negative terminal, $p = -vi$



+Power absorbed = -Power supplied

Figure 1.8

Reference polarities for power using the passive sign convention: (a) absorbing power, (b) supplying power.

Example: Power

- Q: Given the two diagrams shown below, determine whether the element is absorbing or supplying power and how much*

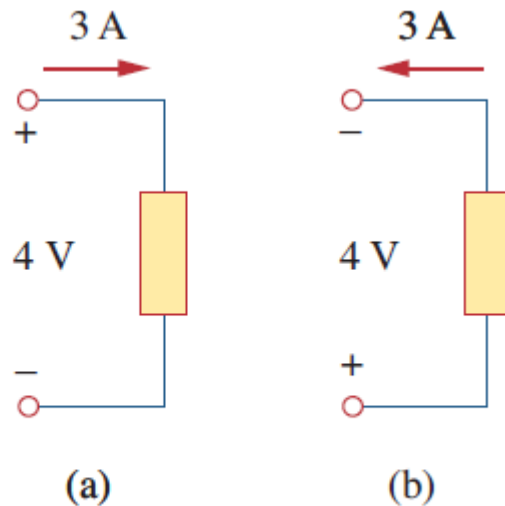


Figure 1.9

Two cases of an element with an absorbing power of 12 W: (a) $p = 4 \times 3 = 12$ W, (b) $p = 4 \times 3 = 12$ W.

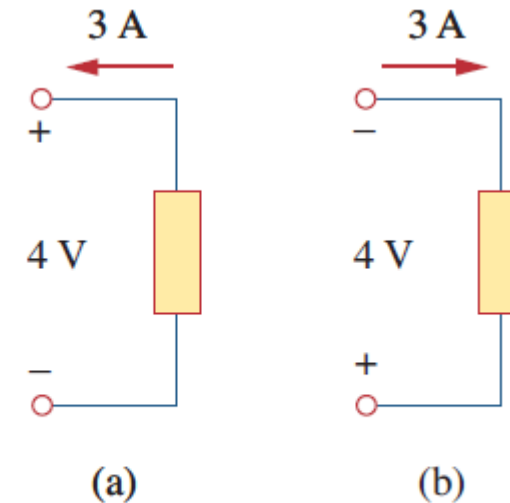


Figure 1.10

Two cases of an element with a supplying power of 12 W: (a) $p = -4 \times 3 = -12$ W, (b) $p = -4 \times 3 = -12$ W.

Power and Energy (IV)

- **Principle of Conservation of the Power (能量守恒定律):** The algebraic sum of the powers absorbed by all elements in a circuit is zero at any instance of time ($\sum P=0$). That is, the sum of absorbed powers is equal to the sum of generated powers at each instance of time

$$\sum p = 0$$

- Energy is the capacity to do work, measured in **joules (J)**. The energy absorbed or supplied by an element from time t_0 to time t is

$$w = \int_{t_0}^t p \, dt = \int_{t_0}^t v i \, dt$$

- The electric power utility companies measure energy in **watt-hours (Wh)**, where $1 \text{ Wh} = 3,600 \text{ J}$ and $1 \text{ kWh} = 1 \text{ 度}$

Circuit Elements

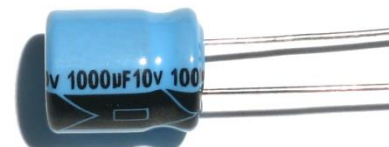
- An **element (元件)** is the basic building block of a circuit. An electric circuit is simply an interconnection of the elements
 - Circuit analysis is the process of determining voltages across (or the currents through) the elements of the circuit
- **Active elements (有源元件)** can generate energy

- Generators
- Batteries
- Operational Amplifiers



- **Passive elements (无源元件)** absorb energy

- Resistors
- Capacitors
- Inductors



(Ideal) Voltage and Current Sources

- The most important active elements are **voltage or current sources** (电压源/电流源) that generally deliver power to the circuit connected to them
- Independent sources** (独立源): An (ideal) independent source is an active element that provides a specified voltage or current that is independent of other circuit elements and/or how the source is used in the circuit
 - **An ideal voltage source has no internal resistance** → we can know the voltage at its terminals, but we don't know in advance the current
 - **An ideal current source has infinite resistance** → We can know the current through it in advance, but not the voltage

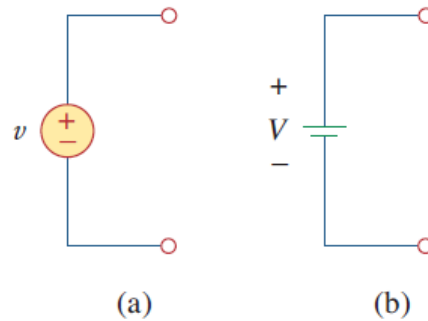


Figure 1.11
 Symbols for independent voltage sources:
 (a) used for constant or time-varying voltage, (b) used for constant voltage (dc).

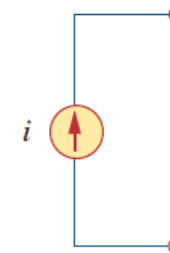


Figure 1.12
 Symbol for independent current source.

Ideal Dependent (Controlled) Source (I)

- An **ideal dependent (or controlled) source (受控源)** is an active element in which the source quantity is controlled by another voltage or current
 - Dependent sources are usually designated by diamond-shaped symbols
- Four types:
 - A voltage-controlled voltage source (VCVS)
 - A current-controlled voltage source (CCVS)
 - A voltage-controlled current source (VCCS)
 - A current-controlled current source (CCCS)

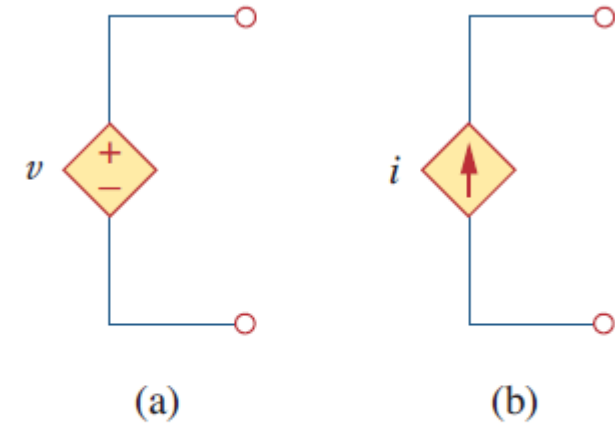
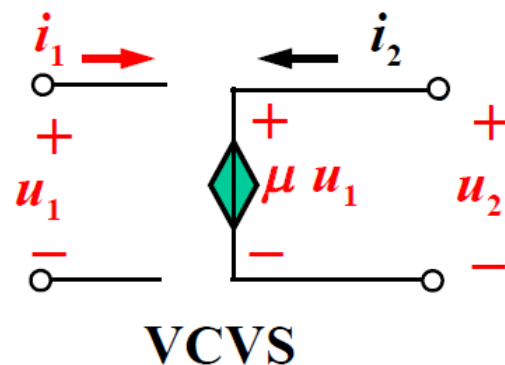


Figure 1.13

Symbols for: (a) dependent voltage source, (b) dependent current source.

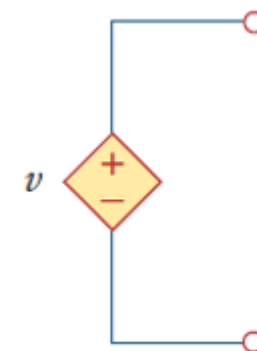
Ideal Dependent (Controlled) Source (II)

- A voltage-controlled voltage source (VCVS, 压控电压源)

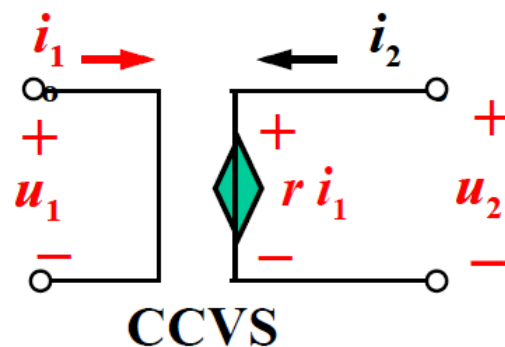


$$\begin{cases} i_1 = 0 \\ u_2 = \mu u_1 \end{cases}$$

μ : 转移电压比



- A current-controlled voltage source (CCVS, 流控电压源)

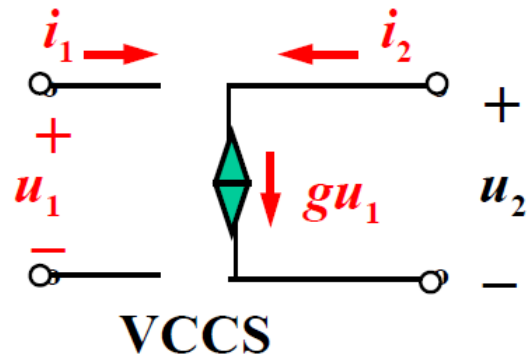


$$\begin{cases} u_1 = 0 \\ u_2 = r i_1 \end{cases}$$

r : 转移电阻

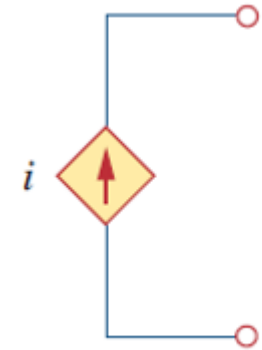
Ideal Dependent (Controlled) Source (III)

- A voltage-controlled current source (VCCS, 压控电流源)

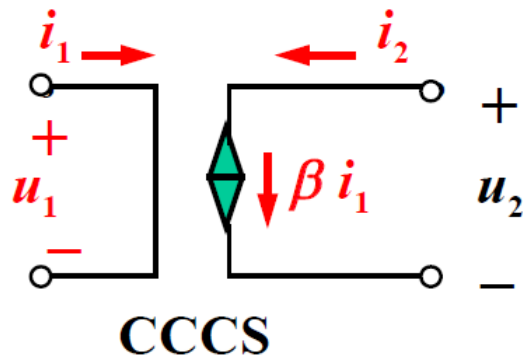


$$\begin{cases} i_1 = 0 \\ i_2 = g u_1 \end{cases}$$

g : 转移电导



- A current-controlled current source (CCCS, 流控电流源)

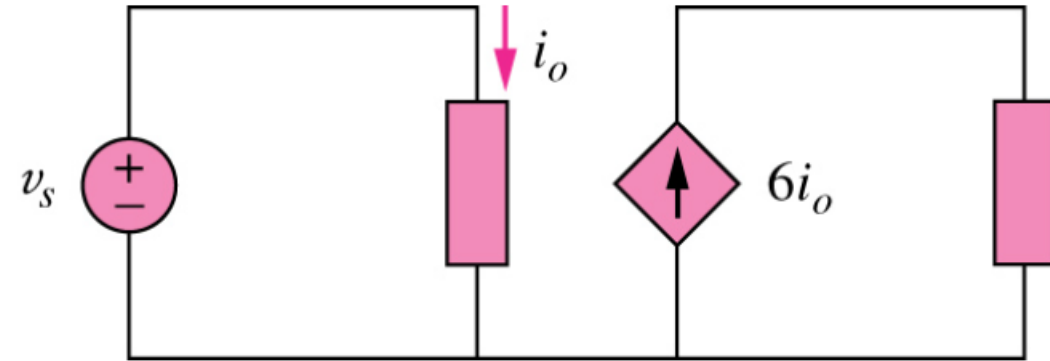
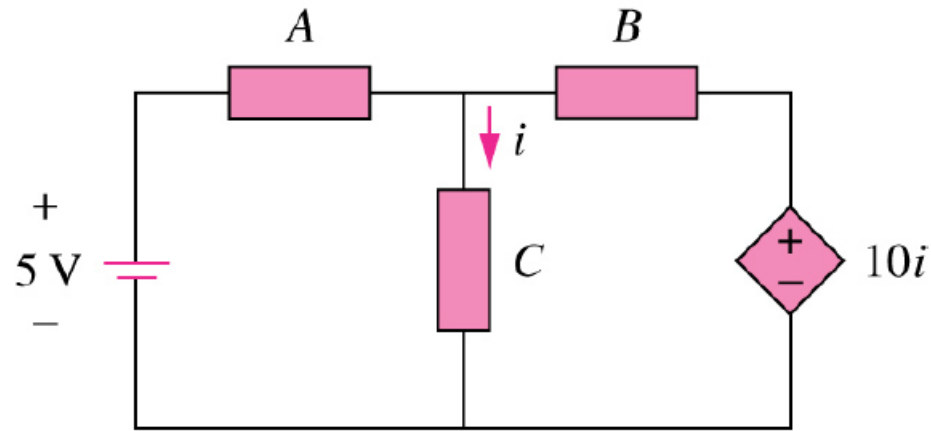


$$\begin{cases} u_1 = 0 \\ i_2 = \beta i_1 \end{cases}$$

β : 转移电流比

Example: Dependent Source

- In the following circuits, identify the type of dependent sources



Example: Power Calculation

Example 1.7

Calculate the power supplied or absorbed by each element in Fig. 1.15.

Solution:

We apply the sign convention for power shown in Figs. 1.8 and 1.9. For p_1 , the 5-A current is out of the positive terminal (or into the negative terminal); hence,

$$p_1 = 20(-5) = -100 \text{ W} \quad \text{Supplied power}$$

For p_2 and p_3 , the current flows into the positive terminal of the element in each case.

$$p_2 = 12(5) = 60 \text{ W} \quad \text{Absorbed power}$$

$$p_3 = 8(6) = 48 \text{ W} \quad \text{Absorbed power}$$

For p_4 , we should note that the voltage is 8 V (positive at the top), the same as the voltage for p_3 , since both the passive element and the dependent source are connected to the same terminals. (Remember that voltage is always measured across an element in a circuit.) Since the current flows out of the positive terminal,

$$p_4 = 8(-0.2I) = 8(-0.2 \times 5) = -8 \text{ W} \quad \text{Supplied power}$$

We should observe that the 20-V independent voltage source and 0.2I dependent current source are supplying power to the rest of the network, while the two passive elements are absorbing power. Also,

$$p_1 + p_2 + p_3 + p_4 = -100 + 60 + 48 - 8 = 0$$

In agreement with Eq. (1.8), the total power supplied equals the total power absorbed.

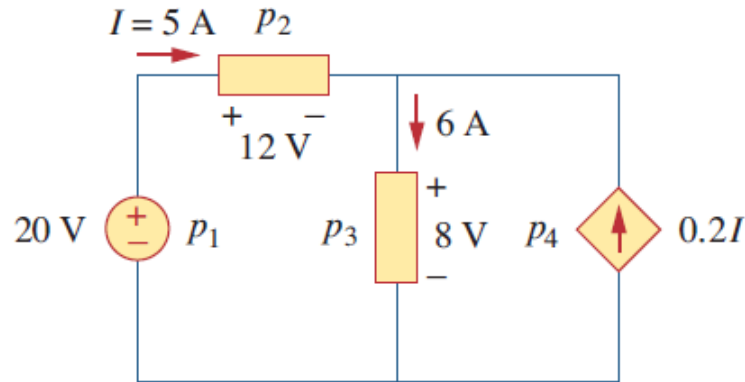


Figure 1.15
For Example 1.7.

Summary

- An electric circuit consists of electrical elements connected together
- Current is the rate of charge flow past a given point in a given direction
- Voltage is the energy required to move a unit charge through an element
- Power is the energy supplied or absorbed per unit time. It is also the product of voltage and current
- According to the passive sign convention, power assumes a positive (negative) sign when the current enters the positive (negative) polarity of the voltage across an element
- An ideal voltage source produces a specific potential difference across its terminals regardless of what is connected to it. An ideal current source produces a specific current through its terminals regardless of what is connected to it
- Voltage and current sources can be dependent or independent

Acknowledgement: This Lecture Note is partially based on materials from Prof. Chenchang Zan and partially adapted from the notes of the book “Fundamentals of Electric Circuits” by C. K. Alexander and M. N. O. Sadiku